NEWTON SCHOOL OF TECHNOLOGY

Data Visualization and Analytics - Capstone 2

ZOMATO DELIVERY OPERATIONS ANALYTICS

Sector: Food Technology and Delivery Operations

Field	Details
Section	C
Team ID	G16
Project Title	Zomato Delivery Operations Analytics
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Institute	Newton School of Technology
GitHub Repository	https://github.com/kushpuri07/C_G16_Zomato-Data-Analysis
Tableau Dashboard	https://public.tableau.com/views/Zomato_Delivery_Analysis/Dashboard1

1. Executive Summary

This report presents the findings of a comprehensive data analytics study conducted on Zomato delivery operations in India. The project was executed as part of the DVA Capstone 2 program at Newton School of Technology by Team G16, Section C.

The primary objective of this project was to identify the operational factors that most significantly drive delivery delays on Zomato and to derive actionable recommendations that can improve on-time delivery rates and overall customer satisfaction.

The dataset comprised 45,584 row-level delivery records spanning multiple Indian cities, with 20 operational attributes including delivery partner demographics, vehicle types, weather conditions, road traffic density, order types, and delivery timestamps. The dataset was sourced from Kaggle and contained realistic data quality issues including missing values, inconsistent formats, and typographic errors, which were resolved through a documented Python ETL pipeline.

Key findings indicate that road traffic density, weather conditions, and the occurrence of festival periods are statistically significant drivers of delivery time. Deliveries during jam traffic conditions take on average 5 to 7 minutes longer than during low traffic periods. Cloudy weather emerged as the weather condition most associated with elevated delivery times. Partners with higher ratings consistently demonstrated lower average delivery times, suggesting that partner quality directly impacts operational efficiency.

Three interactive Tableau dashboards were developed to present these findings across executive, operational, and partner performance dimensions. The analysis concludes with five data-backed business recommendations targeting traffic routing, weather-based staffing, partner incentivisation, and festival-period resource planning.

2. Sector Context and Problem Statement

2.1 Sector Overview

The food technology and on-demand delivery sector in India has experienced exponential growth over the past decade. Platforms such as Zomato and Swiggy have transformed consumer expectations around food delivery, with customers increasingly demanding faster, more reliable service. India's food delivery market is projected to grow at a compound annual growth rate exceeding 15 percent through 2027, making operational efficiency a critical competitive differentiator.

Delivery time is one of the most directly observable measures of operational performance in this sector. Delays caused by traffic congestion, adverse weather, under-resourced delivery partners, or festival-period surges directly translate into customer dissatisfaction, negative ratings, and order cancellations. Understanding the root causes of these delays is therefore of strategic importance.

2.2 Problem Statement

> *Which operational factors — traffic density, weather conditions, vehicle type, delivery partner ratings, and festival periods — most significantly impact food delivery time on Zomato, and what targeted interventions can reduce average delivery time by at least 15 percent?*

2.3 Project Objectives

- Identify and quantify the impact of traffic, weather, and festival conditions on delivery time.
- Analyse delivery partner performance across ratings, age groups, and vehicle types.
- Compute actionable KPIs that enable operational and executive decision-making.
- Build interactive Tableau dashboards for three distinct stakeholder audiences.
- Deliver data-backed business recommendations with estimated impact.

3. Data Description

3.1 Dataset Source

The primary dataset was sourced from Kaggle: Zomato Delivery Operations Analytics Dataset by Saurabh Badole. The dataset is a synthetic simulation of real-world Zomato delivery operations, designed to reflect realistic operational patterns and data quality issues.

DatasetURL:
https://www.kaggle.com/datasets/saurabhbadole/zomato-delivery-operations-analytics-dataset

3.2 Dataset Structure

Attribute	Description	Type
ID	Unique order identifier	Categorical
Delivery_person_ID	Unique delivery partner identifier	Categorical
Delivery_person_Age	Age of the delivery partner in years	Numerical
Delivery_person_Ratings	Customer rating given to the delivery partner	Numerical
Restaurant_latitude	Geographic latitude of the restaurant	Numerical
Restaurant_longitude	Geographic longitude of the restaurant	Numerical
Delivery_location_latitude	Geographic latitude of the delivery location	Numerical
Delivery_location_longitude	Geographic longitude of the delivery location	Numerical
Order_Date	Date on which the order was placed	Date
Time_Orderd	Time at which the order was placed	Time
Time_Order_picked	Time at which the order was picked up	Time
Weather_conditions	Prevailing weather at time of delivery	Categorical
Road_traffic_density	Traffic density level during delivery	Categorical
Vehicle_condition	Condition rating of the delivery vehicle	Numerical
Type_of_order	Category of food ordered	Categorical
Type_of_vehicle	Type of vehicle used for delivery	Categorical
multiple_deliveries	Number of simultaneous deliveries	Numerical
Festival	Whether a festival was ongoing at time of order	Categorical
City	City type of delivery location	Categorical
Time_taken (min)	Total delivery time in minutes (target variable)	Numerical

3.3 Dataset Size

- Total rows: 45,584
- Total columns: 20 (original) | 25 (post feature engineering)
- Sector: Food Technology and Delivery Operations

3.4 Data Limitations

- The dataset is synthetically generated and may not capture all real-world edge cases present in live Zomato operations.
- Geographic coordinates correspond to a limited set of city clusters rather than a fully distributed national sample.
- The dataset does not include customer-side attributes such as order value, cuisine type, or customer loyalty tier.
- Time columns contained a mix of HH:MM string format and decimal representations, requiring custom parsing logic.

4. Data Cleaning and ETL Methodology

4.1 ETL Pipeline Overview

The data engineering pipeline was implemented entirely in Python using Jupyter Notebooks, following the folder structure mandated by the project handbook. The pipeline consisted of two primary notebooks: 01_extraction.ipynb for data loading and initial inspection, and 02_cleaning.ipynb for all transformation and feature engineering steps.

4.2 Extraction - Notebook 01

- Loaded raw CSV using pandas read_csv into a DataFrame.
- Confirmed dataset dimensions: 45,584 rows and 20 columns.
- Inspected data types, column names, and missing value distribution.
- Identified 8 columns with missing values ranging from 228 to 1,908 null entries.
- Identified typographic errors in column names and categorical values.

4.3 Cleaning - Notebook 02

Step 1 — Column Name Correction

The column Time_Orderd contained a typographic error and was renamed to Time_Ordered for consistency.

Step 2 - Categorical Value Correction

The City column contained the misspelling Metropolitian which was corrected to Metropolitan using string replacement.

Step 3 - Missing Value Imputation

Column	Missing Count	Strategy
Delivery_person_Ratings	1,908	Median imputation
Delivery_person_Age	1,854	Median imputation
Time_Ordered	1,731	Mode imputation
City	1,200	Mode imputation
multiple_deliveries	993	Median imputation
Weather_conditions	616	Mode imputation
Road_traffic_density	601	Mode imputation
Festival	228	Mode imputation

Step 4 - Data Type Conversion

- Order_Date was converted from string to datetime using pd.to_datetime with dayfirst=True.
- Time columns were retained as strings for downstream feature extraction.

Step 5 - Feature Engineering

New Feature	Derivation	Purpose
Order_Day	Extracted from Order_Date	Day-level trend analysis
Order_Month	Extracted from Order_Date	Monthly seasonality analysis
Order_Year	Extracted from Order_Date	Year-over-year comparison
Day_of_Week	Extracted from Order_Date using dt.day_name()	Weekly pattern analysis
Order_Hour	Extracted from Time_Ordered using custom parser	Hourly demand analysis

The cleaned dataset was saved to data/processed/Zomato_Cleaned.csv. Post-cleaning, the dataset contained zero null values across all 25 columns.

5. KPI and Metric Framework

The KPI framework was designed to directly address the problem statement. Each KPI was selected to provide decision-relevant insight for one of three stakeholder groups: executive management, operations teams, and partner performance managers.

KPI	Definition	Stakeholder
Average Delivery Time	Mean of Time_taken (min) across all orders	Executive
Total Orders	Count of unique order IDs	Executive
Average Partner Rating	Mean of Delivery_person_Ratings	Executive
Peak Order Day	Day of week with highest order volume	Executive
Worst Traffic Density	Traffic level with highest average delivery time	Operations
Worst Weather Condition	Weather type with highest average delivery time	Operations
Festival Time Difference	Difference in avg delivery time: Festival vs Non-Festival	Operations
Average Multiple Deliveries	Mean number of simultaneous deliveries per order	Operations
Average Partner Age	Mean age of all delivery partners	Partner Performance
Top Vehicle Used	Vehicle type with highest order volume	Partner Performance
Top Rated Average Time	Avg delivery time for partners with rating >= 4.8	Partner Performance

6. Exploratory Data Analysis

6.1 Delivery Time Distribution

The target variable Time_taken (min) follows an approximately uniform distribution ranging from 10 to 54 minutes, with a mean of 26.29 minutes. The distribution shows no significant skewness, indicating that delivery times are broadly spread across the range rather than concentrated at any single value. This distribution pattern suggests that operational variability is high and driven by multiple contributing factors rather than a single dominant cause.

6.2 City-wise Analysis

Semi-Urban areas recorded the highest average delivery time at approximately 49 minutes, followed by Metropolitan areas at approximately 29 minutes, and Urban areas at approximately 20 minutes. This finding is counterintuitive and likely reflects the lower road infrastructure density in semi-urban areas, the fewer available delivery partners, and longer travel distances between restaurants and delivery locations.

6.3 Weather Impact

Cloudy weather conditions were associated with the highest average delivery time, followed by Fog and Windy conditions. Sunny weather was associated with the lowest delivery times. The elevated delivery times under cloudy conditions may reflect reduced visibility and more cautious driving behaviour among delivery partners.

6.4 Traffic Density Impact

Jam traffic conditions produced the highest average delivery times, followed by High, Medium, and Low density in descending order. This relationship is linear and expected. The difference between Jam and Low traffic conditions represents a meaningful operational gap that can be partially addressed through dynamic routing and order batching strategies.

6.5 Festival Period Analysis

Festival orders showed a delivery time difference of approximately 6.3 minutes compared to non-festival orders. Interestingly, the analysis revealed that festival orders were completed faster on average, which may be attributable to increased staffing levels deployed by Zomato during high-demand festival periods.

6.6 Vehicle Type Analysis

The treemap analysis of vehicle types revealed that motorcycles and scooters account for the majority of deliveries. Bicycles, while fewer in number, showed the highest average delivery time, which is consistent with their lower speed. Electric scooters showed the lowest average delivery time among all vehicle types, suggesting they may represent an efficient and scalable vehicle option for urban delivery operations.

6.7 Peak Demand Patterns

The hourly order volume analysis revealed two clear demand peaks: a lunch peak around 13:00 to 15:00 hours and a dinner peak around 19:00 to 22:00 hours. Wednesday was identified as the peak order day of the week. These patterns provide a basis for time-based staffing and resource allocation recommendations.

6.8 Partner Ratings

The scatter plot analysis of partner ratings against delivery time showed a weak but discernible negative correlation: partners with higher ratings tended to deliver orders slightly faster. Partners rated 4.8 and above showed an average delivery time of 24.39 minutes, compared to the overall average of 26.29 minutes.

7. Statistical Analysis

7.1 Methodology

Statistical analysis was performed in notebook 04_statistical_analysis.ipynb using Python's scipy.stats library. Three inferential techniques were applied: One-Way ANOVA for multi-group comparisons, Independent Samples T-Test for two-group comparisons, and Pearson Correlation for numerical relationships. Linear regression was additionally used to model the predictive relationship between operational features and delivery time.

7.2 Hypothesis Tests

Test	Variables	Method	Result
Test 1	Traffic Density vs Delivery Time	One-Way ANOVA	Statistically significant ($p < 0.05$)
Test 2	Festival vs Delivery Time	Independent T-Test	Statistically significant ($p < 0.05$)
Test 3	Weather Conditions vs Delivery Time	One-Way ANOVA	Statistically significant ($p < 0.05$)
Test 4	Multiple Deliveries vs Delivery Time	One-Way ANOVA	Statistically significant ($p < 0.05$)

All four hypothesis tests returned p-values below the significance threshold of 0.05, confirming that traffic density, weather conditions, festival periods, and the number of simultaneous deliveries each have a statistically significant effect on delivery time.

7.3 Pearson Correlation

Pearson correlation coefficients were computed between each numerical feature and the target variable Time_taken (min). Multiple deliveries showed the strongest positive correlation with delivery time. Vehicle condition showed a weak negative correlation, suggesting that better-maintained vehicles are associated with marginally faster deliveries. Partner ratings showed a weak negative correlation, consistent with the EDA findings.

7.4 Linear Regression

A multiple linear regression model was trained on six features: road traffic density, weather conditions, city, multiple deliveries, partner ratings, and vehicle condition. Categorical features were encoded using LabelEncoder prior to modelling. The dataset was split into 80 percent training and 20 percent test sets. The model returned a moderate R-squared score, indicating that these six features collectively explain a meaningful portion of the variance in delivery time. The Mean Absolute Error confirmed the model's practical utility as a baseline predictor for delivery time estimation.

8. Dashboard Design

8.1 Design Principles

All three dashboards were built on Tableau Public following a consistent Zomato-themed design language. The colour scheme employs shades of red (#E24B4A, #A32D2D, #F09595) consistent with the Zomato brand identity. A dark sidebar navigation panel provides seamless switching between dashboards. All KPI cards follow a white background with red top-border treatment. All charts use interactive filters to enable drill-down analysis.

Tableau Public URL: https://public.tableau.com/views/Zomato_Delivery_Analysis/Dashboard1

8.2 Dashboard 1 — Executive Overview

This dashboard provides a high-level summary of overall delivery operations performance. It is designed for senior management and executive stakeholders who require a rapid understanding of key performance metrics without operational detail.

Component	Chart Type	Insight Delivered
Avg Delivery Time	KPI Card	Overall operational baseline
Total Orders	KPI Card	Volume of operations
Avg Partner Rating	KPI Card	Service quality indicator
Peak Order Day	KPI Card	Demand planning anchor
Delivery Locations Map	Geo Map	Geographic spread of operations across India
Avg Time by City	Bar Chart	City-wise performance comparison
Order Type Distribution	Donut Chart	Order category mix
Orders by Hour	Line Chart	Intraday demand pattern
Orders by Day of Week	Bar Chart	Weekly demand pattern

8.3 Dashboard 2 — Operational Analysis

This dashboard is designed for operations managers and logistics teams. It focuses on identifying and quantifying the causes of delivery delays across traffic, weather, festival, vehicle, and order-type dimensions.

Component	Chart Type	Insight Delivered

Worst Traffic Density	KPI Card	Primary traffic bottleneck
Worst Weather Condition	KPI Card	Weather risk indicator
Festival Time Difference	KPI Card	Festival operational impact
Avg Multiple Deliveries	KPI Card	Batching load indicator
Time Taken by Traffic	Donut Chart	Traffic density distribution of delivery time
Avg Time by Weather	Bar Chart	Weather condition impact ranking
Festival Impact	Bar Chart	Festival vs non-festival comparison
Multiple Deliveries Impact	Area Chart	Trend of time increase with batching
Avg Time by Vehicle	Treemap	Vehicle type performance comparison
Avg Time by Order Type	Bar Chart	Order category delivery efficiency

8.4 Dashboard 3 — Partner Performance

This dashboard is designed for partner management and human resources teams. It focuses on delivery partner demographics, performance, and the relationship between partner quality and delivery efficiency.

Component	Chart Type	Insight Delivered
Avg Partner Age	KPI Card	Workforce demographic profile
Avg Partner Rating	KPI Card	Overall service quality
Top Vehicle Used	KPI Card	Fleet composition indicator
Top Rated Avg Time	KPI Card	Best-in-class delivery benchmark
Ratings vs Delivery Time	Scatter Plot	Correlation between quality and speed
Avg Time by City	Bar Chart	Geographic performance variation
Avg Time by Vehicle Condition	Donut Chart	Vehicle maintenance impact
Avg Time by Age Group	Bar Chart	Age-based performance segmentation
Avg Time by Vehicle and Weather	Grouped Bar Chart	Combined vehicle-weather interaction

9. Key Insights

The following insights are expressed in decision language, directly linking analytical findings to operational implications.

No.	Insight
1	Jam traffic conditions increase average delivery time by approximately 5 to 7 minutes compared to low traffic conditions, representing the single largest controllable contributor to delivery delays.
2	Semi-Urban areas record an average delivery time of 49 minutes, nearly double the Urban average of 20 minutes, indicating a structural under-investment in delivery infrastructure and partner density in semi-urban zones.
3	Cloudy weather is associated with the highest average delivery times, suggesting that reduced visibility has a greater operational impact than precipitation-based conditions such as storms or sandstorms.
4	Delivery partners rated 4.8 and above deliver orders an average of 1.9 minutes faster than the fleet average, confirming that partner quality is a measurable and actionable driver of operational efficiency.
5	Each additional simultaneous delivery is associated with a measurable increase in total delivery time, indicating that indiscriminate order batching beyond a threshold reduces both efficiency and customer satisfaction.
6	Festival periods do not consistently increase delivery times when staffing is proactively scaled, as evidenced by the negative festival time difference observed in the data.
7	Electric scooters demonstrate lower average delivery times than motorcycles and bicycles, suggesting they represent a high-value fleet investment for urban delivery operations.
8	Peak order hours at lunch (13:00 to 15:00) and dinner (19:00 to 22:00) are consistent and predictable, enabling precise time-based partner deployment strategies.
9	Delivery partners in the 30 to 35 age group demonstrate the highest average delivery times, while under-25 partners show the lowest, suggesting that younger partners may be more route-adaptive.
10	Vehicle condition has a measurable negative correlation with delivery time: partners operating vehicles in poor condition take approximately 2.76 minutes longer on average than those in good condition.
11	Motorcycle and scooter deliveries account for the majority of all orders, while bicycle deliveries represent a small but consistently slower segment that may warrant route-specific optimisation.
12	Wednesday records the highest order volume across the week, contradicting the common assumption that weekends drive peak demand and suggesting mid-week promotional and staffing adjustments are warranted.

10. Business Recommendations

Recommendation 1 - Implement Dynamic Traffic-Based Routing

Integrate real-time traffic data into the Zomato dispatch algorithm to automatically re-route delivery partners away from jam-density zones during peak hours. Based on the observed 5 to 7 minute delay associated with jam conditions, this intervention could reduce average delivery time by an estimated 8 to 12 percent for orders placed during peak traffic periods.

Recommendation 2 - Deploy Weather-Triggered Surge Staffing

Establish a weather monitoring protocol that automatically triggers additional partner deployment when cloudy or foggy conditions are forecast. Given that cloudy weather is the single most delay-associated weather condition, proactive staffing during these conditions could reduce weather-related delays by an estimated 15 percent.

Recommendation 3 - Introduce Partner Performance Incentive Tiers

Create a tiered incentive structure that rewards delivery partners with ratings of 4.8 and above with priority order allocation and performance bonuses. Given that high-rated partners deliver approximately 1.9 minutes faster, expanding this cohort from current levels to 30 percent of the active fleet could reduce the fleet-wide average delivery time by approximately 0.6 minutes, with compounding customer satisfaction benefits.

Recommendation 4 - Enforce a Multiple Delivery Cap

Limit simultaneous deliveries to a maximum of two orders per partner during peak hours. The data confirms a linear increase in delivery time with each additional simultaneous delivery. Implementing this cap during the 13:00 to 15:00 and 19:00 to 22:00 windows would protect delivery time consistency during the highest-demand periods at a manageable cost to throughput.

Recommendation 5 - Prioritise Semi-Urban Infrastructure Investment

The 29-minute delivery time gap between Semi-Urban and Urban areas represents the largest addressable performance disparity in the dataset. Zomato should prioritise increasing partner density, dark kitchen placement, and route coverage in semi-urban zones. A 20 percent increase in semi-urban partner density is estimated to reduce semi-urban average delivery times by approximately 10 minutes, bringing them closer to Urban benchmarks.

11. Impact Estimation, Limitations, and Future Scope

11.1 Impact Estimation

Recommendation	Estimated Impact
Dynamic Traffic Routing	8 to 12 percent reduction in avg delivery time during peak traffic hours
Weather-Triggered Surge Staffing	15 percent reduction in weather-related delays
Partner Incentive Tiers	0.6 minute reduction in fleet-wide average delivery time
Multiple Delivery Cap	Consistent delivery time protection during peak hours
Semi-Urban Infrastructure	Approximately 10 minute reduction in semi-urban average delivery time

11.2 Limitations

- The dataset is synthetically generated. While it reflects realistic operational patterns, findings should be validated against live Zomato transactional data before implementation.
- The dataset covers a limited time window and may not capture seasonal variation across all quarters of the year.
- Customer-side data such as order value, cuisine preference, and satisfaction scores were not available, limiting the scope of demand-side analysis.
- The linear regression model produced a moderate R-squared score, indicating that a significant portion of delivery time variance is explained by factors not present in the dataset.

11.3 Future Scope

- Integration of real-time GPS and traffic API data to enable live delivery time prediction.
- Application of advanced machine learning models including Random Forest and XGBoost to improve delivery time prediction accuracy.
- Customer churn analysis linking delivery delays to repeat order behaviour and rating outcomes.
- Expansion of the geographic scope to include tier-2 and tier-3 Indian cities for a nationally representative analysis.

12. Contribution Matrix

The following matrix documents the primary contributions of each team member. GitHub Insights, pull request history, and committed files serve as the verifiable source of record for all contributions listed below.

Team Member	Role	Primary Contributions
Kush Puri	Project Lead, Visualization Lead	Repository setup, GitHub branch management, Tableau dashboard design and publishing, Executive Overview and Partner Performance dashboards. Contributed in all the notebooks.
Nitanshu Goyal	Design	Helped in Visualization, listing out KPIs and building the dashboard.
Satyam Singh	Analysis	Visualisations, written insights.Visualization and uploading processed data in github.
Neeraj Singh	Statistical Analysis and Visualization	Looked for dataset, helped in cleaning.Notebook 04 hypothesis testing, correlation analysis, linear regression model
Augustya Purohit	PPT	Presentation deck

GitHub Repository: https://github.com/kushpuri07/C_G16_Zomato-Data-Analysis